EXPLORE PUNE (*A#M/(0@ • 89M/>&M0@)

Complete Cultural Heritage, Sahyadri Trails & Smart AI Tourism Platform

- Interactive Pune Tourism Platform with 111+ Verified Landmarks
- AI Trip Planner with Live Transit & PMPML/Metro Fare Calculation
- Cinematic "My Journey" Travel Storybook with Animated Route Playback
- Milestones, Community Photo Gallery & Dual Feedback Notification Engine
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Presented by: Suraj Dhok & Sahil Jadhav

1. Problem Statement & Opportunity

Addressing fragmented tourism data & lack of intelligent travel guidance

- Fragmented Information: Hours, entry fees, and transit options in Pune are scattered and often outdated.
- Sahyadri Heritage Awareness: Young travelers and trekkers need reliable historical context and safety guidance.
- Missing Personal Travel Diary: No platform preserves a visitor’s day-by-day exploration journey with interactive route maps.
- Opportunity: A unified, authentic, bilingual (English/Marathi) web platform designed specifically for Pune's cultural and geographical landscape.

2. System Architecture & Tech Stack

Modern, performant, and scalable full-stack web architecture

- Frontend Framework: Next.js 16.4.2 (App Router with Turbopack) & React 19
- Styling & Responsiveness: Tailwind CSS v4 + custom responsive system (0 horizontal overflow)
- Backend & Cloud Database: Firebase Cloud Firestore + Firebase Admin SDK
- Authentication: Firebase Auth (Email/Password & Google Sign-In)
- Artificial Intelligence: Google Gemini API (@google/generative-ai) for itinerary planning & chatbot
- Maps & Graphics: MapLibre GL JS + Custom Hand-Crafted SVG Animated Route Engine
- Document & Presentation Export: jsPDF for verified itineraries and milestone certificates

3. Key Features Overview

End-to-end capabilities serving tourists, trekkers, and history enthusiasts

- Explore Engine: 111+ published places across Forts, Heritage, Nature, Temples, and Museums.
- Real-time Operational Status: Dynamic Open/Closed indicators based on verified visiting hours.
- AI Trip Planner: 3 distinct plan options based on budget, travel pace, and start location.
- Integrated Public Transit: Live PMPML pass calculations (₹70 PMC/PCMC, ₹150 All-route) & Metr
- My Journey Storybook: 3-level travel log (Today, Date-wise Timeline, and Overall Conquer Map).
- Community & Milestones: Badges for check-ins, public photo galleries, and printable certificates.

4. "My Journey" Cinematic Travel Experience

Transforming exploration into an interactive digital storybook

- Level 1 — Today’s Journey: Live tracking of current day trips, stop completions, and remaining distance.
- Level 2 — Date-Wise History: Calendar & timeline browsing of past explorations (e.g., 14 June, 20 Oct).
- Level 3 — Overall Conquest: Combined exploration map of Pune with aggregate km traveled and places conquered.
- Animated Route Playback: Smooth Bezier SVG animated curves connecting Pune coordinates with ripple markers.
- Native Audio Pronunciation: Marathi native audio assistance for cultural landmarks.

5. AI Trip Planner & Smart Transit Engine

Grounded algorithmic & LLM intelligence for realistic day tours

- Grounded Context: Gemini API receives authentic Firestore data—eliminating hallucinations.
- Multi-Modal Routing: Incorporates road travel, PMPML bus numbers, and Pune Metro Aqua/Purple lines.
- Cost Estimation: Accurate daily passes (₹70 local / ₹150 full-network) and monument entrance fe
- Guided Tour Mode: Step-by-step navigation allowing real-time place check-ins on active trips.
- Exportable PDF: Generates branded PDF itineraries complete with timings, maps, and transit guidance.

6. Gamification, Community & Feedback

Engaging user retention and crowd-sourced verification

- Milestone Badges: Puneri Explorer, Sahyadri Trekker, Peshwa Heritage, and Cultural Sentinel awards.
- Photo Verification: Upload visit proof with privacy control (Public community photo vs Private personal log).
- Dual-Notification Feedback: Submitting feedback saves to Firestore and triggers dual automated emails:
  - User Confirmation: Personalized polite reply with PuneTRAIL branding.
  - Admin Notification: Instant alert delivered to project leads (Suraj Dhok & Sahil Jadhav).

7. Quality Assurance & Engineering Rigor

Comprehensive audit results ensuring production readiness

- TypeScript Strict Mode: 0 errors across 200+ source files.
- ESLint Standard: 0 warnings and 0 errors.
- Automated Test Suite: 23/23 tests passing with Vitest (Security, Planner, Journey, Distance math).
- Responsive Design: Verified 0 horizontal overflow across all screens (375px mobile to 1920px desktop).
- Hydration Stability: Clean SSR hydration with next/script and suppressHydrationWarning compliance.
- Safe LocalStorage: Corruption-resistant JSON parsing and array validation guards.

8. Deployment & Hosting

Live links and repository access

- Firebase Hosting Site: https://punetrail.web.app
- Alternative Deployment: https://explore-pune.vercel.app (1-Click Vercel GitHub Integration)
- GitHub Source Repository: https://github.com/sj176644-maker/ExplorePune
- Local Production Server: http://localhost:3000
- Continuous Deployment: Auto-build triggers on git push to the main branch.

9. Future Roadmap & Conclusion

Next steps for Pune's premier digital tourism guide

- Offline PWA & Sahyadri Topographic GPS: Offline caching for high-altitude fort treks with weak cellular signal.
- Augmented Reality (AR) Views: 3D architectural reconstructions of historic structures (Shaniwar Wada, Lal Mahal).
- Audio Guide Integration: GPS-triggered audio walking tours narrated in Marathi, English, and Hindi.
- Summary: Explore Pune is a fully functional, production-ready, highly localized digital ecosystem.
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Thank You! Questions & Discussion.